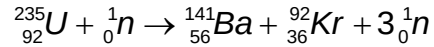


Q6(a)(i)



Q6(a)(ii)

Energy released from an interaction from a U-235 nucleus is

$$E = [(m_U + m_n) - (m_{Ba} + m_{Kr} + 3m_n)]c^2$$

Number of U-235 in $M = 500$ g of U-235 is $N = \frac{M}{M_m} N_A$

$$\begin{aligned} \text{Total energy liberated is } E &= \left(\frac{M}{M_m} N_A \right) [(m_U + m_n) - (m_{Ba} + m_{Kr} + 3m_n)]c^2 \\ &= \left(\frac{500}{235} \times 6.02 \times 10^{23} \right) [(235.1175 + 1.00898) - (140.9577 + 91.9264 + 3 \times 1.00898)] (1.66 \times 10^{-27}) (3.0 \times 10^8)^2 \\ &= \mathbf{4.12 \times 10^{13} \text{ J}} \end{aligned}$$

Q6(a)(iii)

Each U-235 nucleus fissioned by an incoming neutron produces 3 neutrons which can lead to more fissions in a chain reaction in a very short time.

Q6(b)(i)

It is the time taken for the number of radioactive nuclei to decrease of half its original number.

Q6(b)(ii)

At time t or the age of the ore, $\frac{N_U}{N_{Pb}} = \frac{2.8}{1} \dots\dots(1)$

This means that , from $N = N_0 e^{-\lambda t}$ for uranium, we have $\frac{N}{N_0} = \frac{2.8}{3.8}$

$$\frac{2.8}{3.8} = e^{-\left(\frac{\ln 2}{t_{\frac{1}{2}}}\right)t} \text{ or } t = \left(\frac{t_{\frac{1}{2}}}{\ln 2}\right) \ln\left(\frac{3.8}{2.8}\right)$$

$$t = \left(\frac{4.5 \times 10^9}{\ln 2}\right) \left(\ln \frac{3.8}{2.8}\right) = \mathbf{2.0 \times 10^9 \text{ yr}}$$

Q7(a)